

Session 5:

How AI Learns from Data

Now that you know what Machine Learning is and the different ways machines can learn, it's time to dive deeper into the fuel that powers it all - **DATA!**



Objectives

By the end of this session, you will be able to:

01

Understand Datasets

Explain what a dataset is and why it matters

02

Define Features

Explain what AI pays attention to

03

Distinguish Data Types

Tell the difference between training and testing data

04

Explain Memorization vs Learning

Understand why AI must learn patterns, not just memorize

05

Define Generalization

Explain what it means for AI to truly learn

06

Identify Overfitting

Recognize when AI memorizes too much

07

Spot Data Problems

Identify issues with limited or biased data

08

Value Data Quality

Explain why good data matters

Recap: What We Learned in Session 4

Last time, we discovered the different ways machines can learn. Let's refresh our memory:

Concept	Key Idea
Supervised Learning	Learning with labeled examples – like flashcards with answers
Unsupervised Learning	Finding patterns on its own without labels
Reinforcement Learning	Learning by trial and error with rewards and penalties
Real AI Systems	Most AI systems combine multiple types of learning

📌 Today we're diving deep into the **first and most important part of the AI Recipe: DATA!**



Big Question of the Day

What makes good data,
and why does it matter
so much?

The answer isn't simple! Today we'll discover why **data quality**,
quantity, and **variety** all matter.

What is a Dataset?

A **dataset** is a collection of examples used to train an AI. It's like a big box of practice problems for the AI to learn from.

Remember in Session 1, you used Google Storybook to create stories with AI. That AI was trained on a massive dataset of books and stories! And in Session 2, you learned that data is the fuel of AI. Now you'll see what that really means.

AI System	What's in Its Dataset
Spam Filter in Gmail	Millions of emails, labeled "spam" or "not spam"
ChatGPT	Billions of web pages, books, articles – basically the whole internet!
YouTube Recommendations	Millions of videos and data on what people watch and like
Google Storybook	Thousands of children's books and stories

  **Key Idea:** A dataset is just a collection of examples - but the quality of those examples changes everything!

A Tiny Dataset Example

Here's what a small dataset for recognizing fruits might look like:

Picture	Label	Color	Shape	Texture
	Apple	Red	Round	Smooth
	Banana	Yellow	Long	Smooth
	Orange	Orange	Round	Bumpy
	Strawberry	Red	Small, heart	Seeds outside

  **Key Idea:** A dataset is just a collection of examples – but the quality of those examples changes everything!

Features - What AI Pays Attention To

Features are the clues or characteristics that an AI uses to make decisions. They're like the clues a detective uses to solve a mystery.

Remember in Session 3, you learned about pattern recognition - how AI finds patterns in data. Features are the clues the AI looks at to find those patterns!

How It Works : Apples vs. Oranges

Clue (Feature)	Apple	Orange
Color	Red, green, or yellow	Orange
Shape	Round	Round
Texture	Smooth skin	Bumpy skin
Size	Medium	Medium
Has a stem?	Usually yes	Sometimes

The AI looks at these clues and learns patterns: *"If it's orange AND bumpy, it's probably an orange. If it's red AND smooth, it's probably an apple."*

The Detective Analogy

A detective solving a mystery looks for clues:



Color Clue

What color was the car?



Height Clue

How tall was the suspect?



Time Clue

What time did it happen?



 AI does the same thing - it looks at the **clues in the data** to find patterns!



Choosing Good Clues (Features)

Not all clues are equally useful. Some are helpful, some are distractions, and some are totally useless.

Problem	Helpful Clues	Useless Clues
Spam detection	Words in email, sender address	Color of the email background
Face recognition	Distance between eyes, nose shape	What shirt you're wearing
Weather prediction	Temperature, clouds, wind	Your horoscope that day
Movie recommendations	Genre, actors, your past ratings	The color of the movie poster



Key Idea: Choosing the right clues is super important. **Bad clues = bad predictions!**

Training Data vs. Testing Data

When we teach an AI, we don't give it all our data at once. We split it into two groups:

Training Data (80%)

Used to **teach** the AI. Like practice problems before a test. You study these and get better.

Testing Data (20%)

Used to **check** if it learned. Like the actual test. These are NEW problems you've never seen.

Mini Activity 1 : Be a Detective

Part A: Find the Clues

Your mission: Help an AI learn to tell the difference between cats and dogs!

What clues (features) would you want the AI to pay attention to? List at least 4:

Clue	What to Look For in Cats	What to Look For in Dogs
1. Ear shape		
2. Face shape		
3. Sound they make		
4. Tail		

Bonus: What clue would be completely useless? _____

Part B: Split the Data (3 minutes)

You have 100 photos of cats and dogs. Answer these questions:

How many photos for practice (training)? _____

How many photos for the test (testing)? _____

Why can't you use the same photos for both?

What happens if you only test on photos the AI has already seen?

Mini Activity 1: Answer Key

Part A: Find the Clues (Sample Answers)

Clue	Cats	Dogs
Ear shape	Pointed ears	Floppy ears (usually)
Face shape	Rounder, shorter face	Longer snout/muzzle
Sound	Meow, purr	Bark, growl
Tail	Often upright, twitchy	Often wagging

Useless clues: Background color, time of day, what collar they're wearing, if they're inside or outside

Part B: Split the Data (Answers)

Training Data

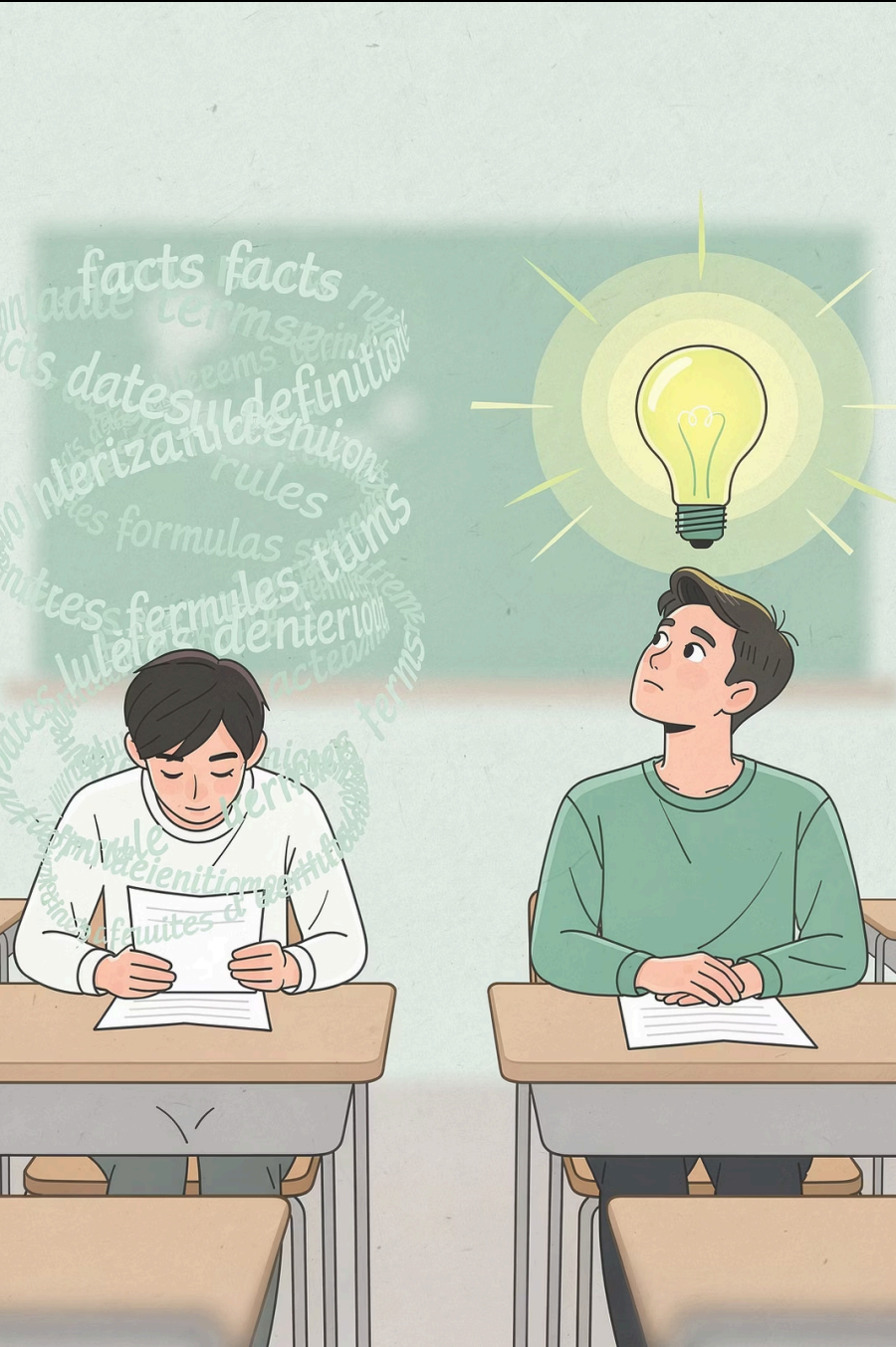
80 photos (80% of 100)

Testing Data

20 photos (20% of 100)

Why separate? If the AI sees the test photos during training, it might just memorize them. We need brand new photos to see if it really learned the concept of "cat" and "dog."

Testing on seen photos: The AI could score 100% just by remembering, but it would fail on real new photos. Like getting 100% on a test because you memorized the answer key, but failing when asked a new question!



Memorization vs. Learning

This is one of the most important ideas in all of AI and it connects to how **YOU** learn!

Memorization	Learning
Remembering the exact answers	Understanding the patterns
Works only for stuff you've seen	Works for NEW stuff
Like memorizing phone numbers	Like understanding how numbers work
Breaks when things change	Adapts to new situations
"I know this because I've seen it"	"I know this because I understand it"

What AI Should Do

What We Want

- AI that LEARNS (understands patterns)
- AI that works on new stuff
- AI that can handle surprises

What We Don't Want

- AI that just MEMORIZES examples
- AI that only works on practice problems
- AI that breaks with new situations

  **Key Idea:** Memorization looks like learning at first, but it fails when something new shows up. True learning means understanding the pattern!

Generalization - The Goal of Learning

Generalization is a fancy word for being able to handle NEW situations. It's what we want AI to do!

Practice Photos

 (poodle photos)

AI Learns: "Dogs have fur, four legs, floppy ears, tails - the **IDEA** of 'dog'"

New Photo

 (golden retriever - never seen!)

AI Says: "That's a dog! It has fur, four legs, a tail – definitely a dog!" 

Situation	Memorization	Generalization
Learning vocabulary	Memorizing word lists	Understanding word roots
Learning math	Memorizing formulas	Understanding why formulas work
Learning a game	Memorizing specific moves	Understanding game strategy

  The real world is full of new stuff AI has never seen. If it only memorized, it would be useless!

Overfitting - When AI Memorizes Too Much

Overfitting happens when AI learns the practice problems TOO WELL - including random stuff that doesn't actually matter. It's what happens when memorization goes too far.

Good Generalization

- You learn: pointy ears, whiskers, tail
- You can recognize **any** cat

Overfitting (Bad Memorization)

- You memorize: the exact couch, the lighting, a scratch on one photo
- You only recognize cats on that exact couch

Why Overfitting Happens

Cause	Explanation
Too few practice examples	Not enough photos to learn the real pattern
Too many useless clues	AI pays attention to stuff that doesn't matter
Practicing too long	AI keeps "studying" until it memorizes everything

Overfitting Example

Practice Photos: 🐱 (cat on red couch) 🐱 (cat on blue rug) 🐱 (cat on wooden bed)

Good AI (Generalization) ✅

"AI learned what cats look like - pointy ears, whiskers, tail. AI can recognize cats"

Overfit AI (Too Much Memorization) ❌

"AI learned cats on red couches, blue rugs, and wooden beds. If a cat is on grass, AI has **no idea** what it is! It also thinks the background color is super important"



Mini Activity 2 : Spot the Difference

Part A: Good Learning or Too Much Memorization?

For each, decide if the AI is **Learning Well (Generalization)** or **Memorizing Too Much (Overfitting)** :

Scenario	Generalization or Overfitting ?
An AI trained on golden retrievers correctly identifies a poodle as a dog	
An AI trained on sunny day driving crashes when it rains	
An AI that recognizes your face even with glasses or a new haircut	
An AI that only knows cats if they're on the exact same couch	
An AI that can read handwriting it's never seen before	

Part B: Fix the Problem

An AI is trained to recognize stop signs - but only ones on straight roads. It fails on stop signs at curved roads.

Question	Your Answer
What's the problem?	
Why did this happen?	
How would you fix it?	

  Have you ever memorized something without understanding it? What happened?

Mini Activity 2: Answer Key

Part A: Good Learning or Too Much Memorization?

Scenario	Answer	Why?
AI trained on golden retrievers correctly identifies a poodle as a dog	Generalization	It understood the concept of "dog" beyond just the specific breed it saw
AI trained on sunny day driving crashes when it rains	Overfitting	It only memorized sunny conditions, didn't learn the general rules of driving
AI that recognizes your face even with glasses or a new haircut	Generalization	It learned the essential features of your face, not surface details
AI that only knows cats if they're on the exact same couch	Overfitting	It memorized the background instead of learning what cats actually look like
AI that can read handwriting it's never seen before	Generalization	It learned the patterns of letters, not specific examples

Part B: Fix the Problem

Question	Sample Answer
What's the problem?	The AI only recognizes stop signs on straight roads. It fails on curved roads because it overfit to the training data.
Why did this happen?	The training data only contained stop signs on straight roads. The AI mistakenly thought "straight road" was part of what makes a stop sign.
How would you fix it?	Add many examples of stop signs on curved roads, different angles, different lighting, different backgrounds to the training data. Also make sure the testing data includes curved roads.

Limited Data - Not Enough Examples

When you don't have enough data, AI can't learn properly. It's like trying to learn a language from only three sentences.

1 photo

AI memorizes that ONE thing. It thinks all cats look exactly the same!

10+ photos

AI starts to see patterns, but might get confused

1,000+ photos

AI can learn real patterns and ignore weird exceptions

1,000,000+ photos

AI becomes efficient!

Biased Data - Unfair Examples

Bias means the data doesn't represent the real world fairly. Some groups, situations, or examples are missing or there are too many of one kind.

Who collected the data?

If only one person took all photos, their choices affect everything

Where were photos taken?

Photos from one country might not work in another

When were they taken?

Old photos might not show what things look like now



Real Examples of Data Problems

Example 1 : Self-driving Car

The Problem: A self-driving car was trained only on videos of sunny day driving.

What Happened: When it rained, the car got completely confused because it had never seen rain before.

Why: The training data lacked variety - it didn't include different weather conditions.

Example 2 : Facial Recognition

The Problem: Early face recognition was trained mostly on light-skinned faces.

What Happened: It worked great for light-skinned people but failed for dark-skinned people.

Why: The training data didn't include enough variety - it was biased.

Example 3 : Hiring

The Problem: Amazon built an AI to screen job applications, trained on 10 years of their own hiring data.

What Happened: The AI learned to prefer men over women because that's what the old data showed.

Why: The data reflected past bias, and the AI learned that bias.



Bad data doesn't just make AI less accurate - it can be **unfair to real people!**

Why Data Quality Matters

Garbage In = Garbage Out (GIGO)

If the data you put in is bad, the AI that comes out will be bad too.

The Four Things Good Data Needs



Enough

Lots of examples -
1,000s photos, not just
10



Clean

Correct labels, no
mistakes - photos
labeled "cat" are actually
cats



Varied

All kinds of examples -
all skin tones, all
weather, all breeds



Fair

No group left out - equal
representation for
everyone

The Cooking Analogy

🔍 Cooking

- Fresh ingredients + good recipe = **yummy meal**
- Spoiled ingredients + good recipe = **gross meal**
- Fresh ingredients + bad recipe = **weird meal**

🤖 AI

- Good data + good algorithm = **smart AI**
- Bad data + good algorithm = **useless AI**
- Good data + bad algorithm = **unpredictable AI**

💡 You can have the best recipe (algorithm) in the world, but if your ingredients (data) are bad, your meal (AI) will be bad too!



Putting It All Together

If your data is...	Then AI might...	Because of...
Limited (too few examples)	Memorize instead of learn	Overfitting
Biased (unfair representation)	Be unfair to certain groups	Bias in data
Low quality (wrong labels)	Make wrong predictions	Garbage In = Garbage Out
Not varied (only one kind)	Fail in new situations	Poor generalization

The Goal : Good Data = Good AI



✓ Enough examples



✓ Correctly labeled



✓ Varied and diverse



✓ Fair to everyone

Hands-On Activity : Create Your Own Dataset

You'll create a dataset for an AI that decides whether food is **Healthy** or **Unhealthy**. This will help you understand what goes into building a dataset.

Step 1: Decide Your Categories

Category	What counts as this?
Healthy	
Unhealthy	

Step 2: List Your Clues (Features)

What clues should the AI look for? List at least 4:

Clue	Why It Matters	Example Values
1.		
2.		
3.		
4.		

Ideas: Sugar content, calories, ingredients, processed vs. whole, fiber, vitamins

Step 3: Plan for Variety

How will you make sure your data has variety and is fair?

Think About	Your Plan
Different foods (pizza, salad, fruit, candy?)	
Different cuisines (Mexican, Chinese, Italian, Indian?)	
Tricky foods (foods that could go either way?)	
Different brands (organic vs regular?)	

Step 4: Create Your Examples

Fill in 6 examples : 3 healthy, 3 unhealthy. Be specific!

Food Item	Important Clues (list 2-3)	Your Label
1.	• •	
2.	• •	
3.	• •	
4.	• •	
5.	• •	
6.	• •	

Step 5: Prevent Overfitting

Look at your dataset. Could an AI memorize instead of generalize? How would you prevent that?

Risk	How to Prevent It

Hands-On Activity: Answers

Step 1 & 2: Categories & Clues

Category	What counts as this?
Healthy	Fresh fruits, vegetables, lean proteins, whole grains, foods with vitamins and minerals
Unhealthy	Foods high in added sugar, saturated fat, processed snacks, sugary drinks, fast food

Clue	Why It Matters	Example Values
Sugar content (grams)	High sugar often means unhealthy	Apple: 19g, Soda: 39g
Processing level	Whole foods are healthier	Apple: whole, Chicken nuggets: processed
Fiber content	High fiber = healthier	Broccoli: high, Candy: none
Ingredients list length	Long list often means processed	Apple: 1 ingredient, Chips: many

Step 3: Plan for Variety

Think About	Your Plan
Different foods	Include fruits (apple, banana), vegetables (broccoli, carrot), fast food (pizza, burger), snacks (chips, candy), drinks (soda, juice), homemade meals (grilled chicken, salad)
Different cuisines	Mexican (tacos, nachos), Chinese (fried rice, dumplings), Italian (pizza, pasta), Indian (curry, samosa), American (burger, hot dog)
Tricky foods	Granola bars (some healthy, some candy bars in disguise), fruit juice (natural sugar but lots of it), salads with creamy dressing, diet soda (zero sugar but artificial)
Different brands	Organic vs regular brands, name brand vs store brand, restaurant vs homemade versions

Step 4: Examples

Food Item	Important Clues	Label
Apple	19g sugar (natural), whole food, 1 ingredient	Healthy
Broccoli	High fiber, vitamins, no processing	Healthy
Grilled chicken	Lean protein, minimal processing	Healthy
Soda	39g sugar (added), no nutrients	Unhealthy
Potato chips	High in saturated fat, highly processed	Unhealthy
Candy bar	High sugar, high fat, long ingredient list	Unhealthy

Step 5: Prevent Overfitting

Risk	How to Prevent It
AI memorizes these 6 specific foods	Add at least 50-100 examples with lots of variety
AI learns that "red = healthy" (because apple is red)	Include healthy foods of all colors (green broccoli, orange carrots)

Summary : What We Learned Today

Dataset

A collection of examples used to teach AI - like a box of practice problems

Feature

A clue that AI uses to make decisions - like a detective's clues

Training Data

Practice problems AI studies to learn

Testing Data

New problems that check if AI really learned

Memorization vs Learning

Memorizing looks good but fails; learning understands patterns

Generalization

Being able to handle NEW situations - understanding the pattern, not just memorizing

Overfitting

When AI memorizes the training data too well and fails on new stuff

Limited Data

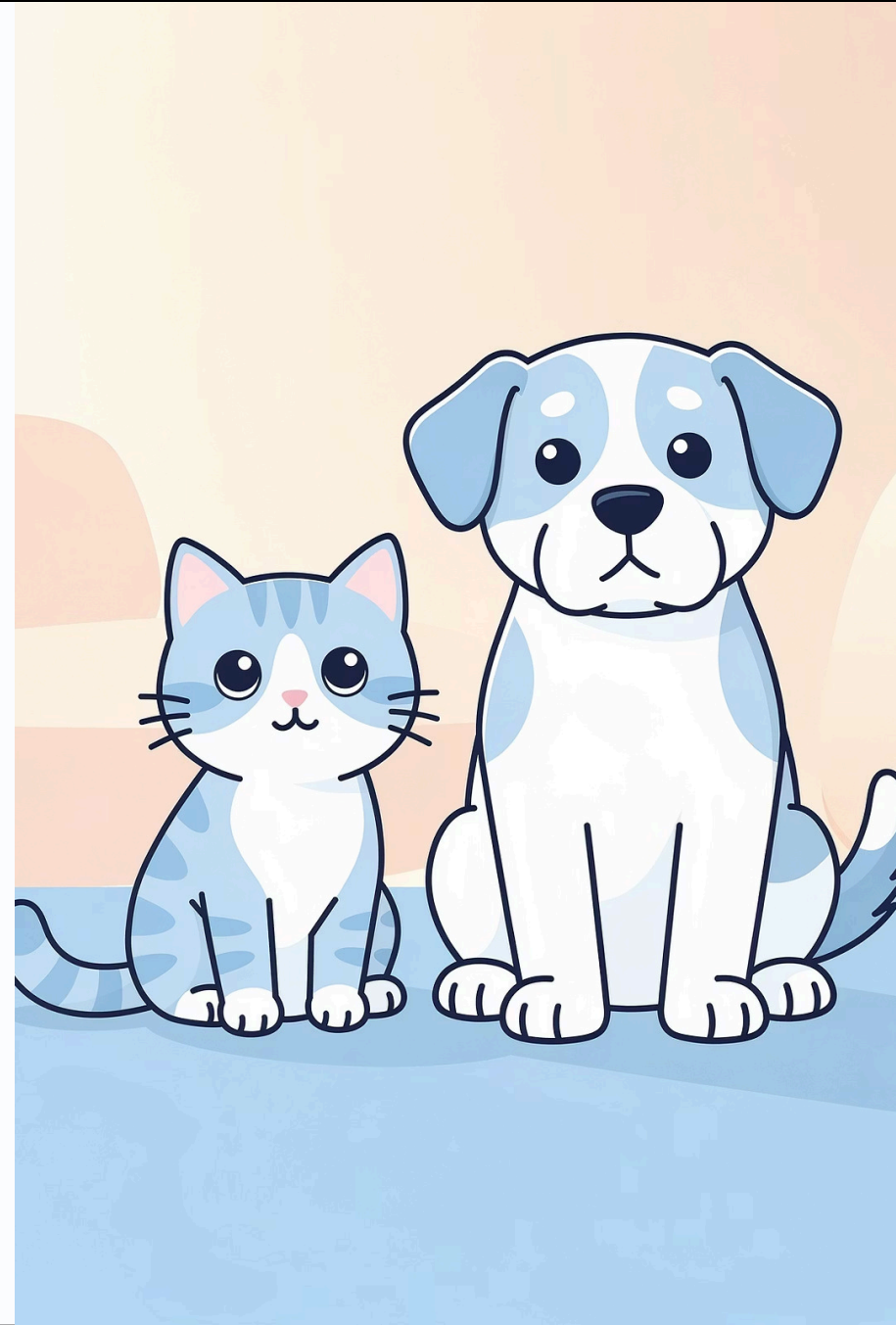
Not enough examples = AI can't learn

Bias Data

When data is unfair - some things are missing or there are too many of one kind

Homework: Prepare for Your Kaggle Project!

In Session 6, you'll build an AI using the famous Cats vs Dogs dataset from Kaggle. Today, you'll plan that project using everything you learned in Session 5.



Step 1: Name Your Classes

Our AI will recognize two classes:

Class 1	Class 2
 Cat	 Dog

Question: What type of Machine Learning is this? *(Hint: We have labeled data!)*

Answer: _____ *(from Session 4)*

Step 2: Predict the Features (Clues)

What features (clues) do you think the AI will use to tell cats and dogs apart? Think about what **YOU** look for.

Feature	CATS	DOGS
Ear shape		
Face shape		
Fur/coat		
Nose		
Tail		
Size		

 Circle the **3 features** you think are MOST important

Step 3: Plan for Variety (Prevent Overfitting!)

Remember: If our data doesn't have variety, the AI might overfit - it could memorize the background instead of learning what cats and dogs actually look like.

What kind of variety?	What should be included?	Your Answer
Breeds	What cat breeds? What dog breeds?	
Colors	What cat colors? What dog colors?	
Backgrounds	Where might the photos be taken?	
Lighting	Bright? Dim? Natural light?	
Angles	Front? Side? From above?	

Step 4: Identify Potential Problems

Based on today's lesson, what could go wrong with our cat vs dog AI?

Problem	Could this happen?	How could we prevent it?
Limited data (not enough examples)		
Bias (some cats/dogs missing)		
Overfitting (memorizing backgrounds)		
Wrong features (AI pays attention to useless clues)		

Next Session Preview

Session 6: Training Your Own AI Model

Now that you understand DATA, it's time to **BUILD** using a real dataset from Kaggle!



Explore Kaggle

Learn about the world's largest dataset community



Load Real Data

Use the famous Cats vs Dogs dataset



Train a Model

Use Teachable Machine to build your first AI



Test It

See how well it works on new images